

Lesson 14 In-Class Worksheet

Reinforcement Learning II: Policy Iteration

Part 1: Instructor-Led Walkthrough

Objective: Apply Policy Iteration to a simple decision problem.

Scenario: An autonomous sentry is stationed at a base (S). It must decide between two actions: *Patrol* or *Recharge*.

- **Action: Patrol (P)** $\rightarrow$ Immediate Reward: +5
- **Action: Recharge (R)** $\rightarrow$ Immediate Reward: 0

Assume the discount factor $\gamma = 0.9$ and both actions loop right back to the base state (S) deterministically.

Step 1: Initialization

We initialize the sentry with a flawed, random policy. Let the initial playbook be:
 $\pi_0(S) = \text{Recharge}$.

Step 2: Policy Evaluation

Calculate the value of the state under this specific playbook (π_0).

Hint: $V^\pi(s) = R(s, \pi(s), s') + \gamma V^\pi(s')$

Step 3: Policy Extraction (Improvement)

Run a one-step lookahead against the environment to find the best immediate action based on V^{π_0} .

Expected utility of Patrol:

Expected utility of Recharge:

New extracted policy $\pi_1(S) =$

Step 4: Convergence Verification

Evaluate $\pi_1(S)$ to find $V^{\pi_1}(S)$, then run extraction again to find $\pi_2(S)$.

Does $\pi_2 = \pi_1$?

Part 2: Student On Their Own

Instructions: Complete the following questions to check your understanding of the lesson objectives.

Question 1: Value Iteration vs. Policy Iteration

While both algorithms solve Markov Decision Processes, they take different operational paths. In your own words, explain the fundamental difference between how Value Iteration and Policy Iteration reach the optimal solution. Which one focuses on the "playbook" at every step?

Question 2: Online vs. Offline Planning

Solving these models with Policy Iteration is considered *Offline Planning*. Explain the key difference between Offline Planning and *Reinforcement Learning* (Online Planning). How does the agent's interaction with the environment differ?

Question 3: Algorithm Exits

In Policy Iteration, how does the AI agent definitively know when to stop running its loops and exit the algorithm?

Question 4: Policy Evaluation Math

An AI commander must calculate the exact expected utility of following an established Standard Operating Procedure (SOP). To do this, the commander uses Policy Evaluation. Why is the $\max_a$ operator removed from the Bellman equation during this specific calculation?